

# 4

## Managing the Database Instance

# Objectives

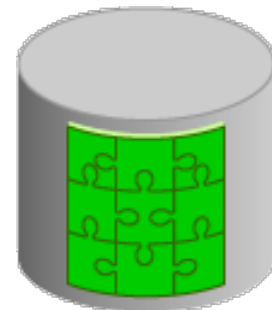
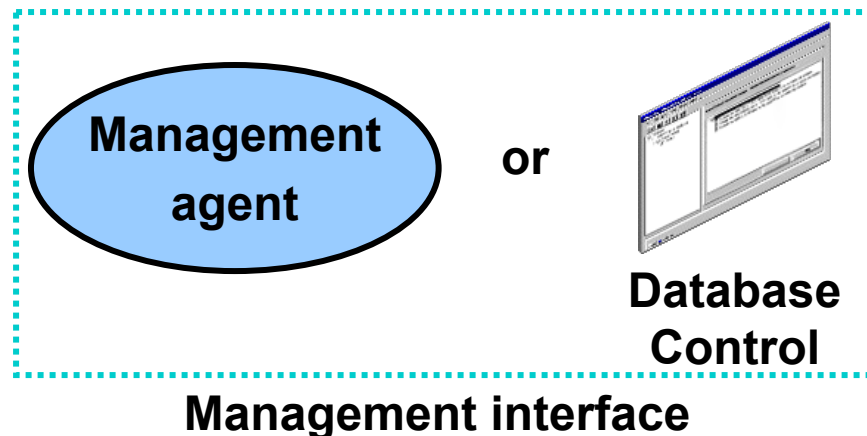
After completing this lesson, you should be able to:

- Start and stop the Oracle database and components
- Use Oracle Enterprise Manager
- Access a database with SQL\*Plus
- Modify database initialization parameters
- Describe the stages of database startup
- Describe database shutdown options
- View the alert log
- Access dynamic performance views

# Management Framework

Oracle Database 11g Release 2 management framework components:

- Database instance
- Listener
- Management interface:
  - Database Control
  - Management agent (when using Grid Control)

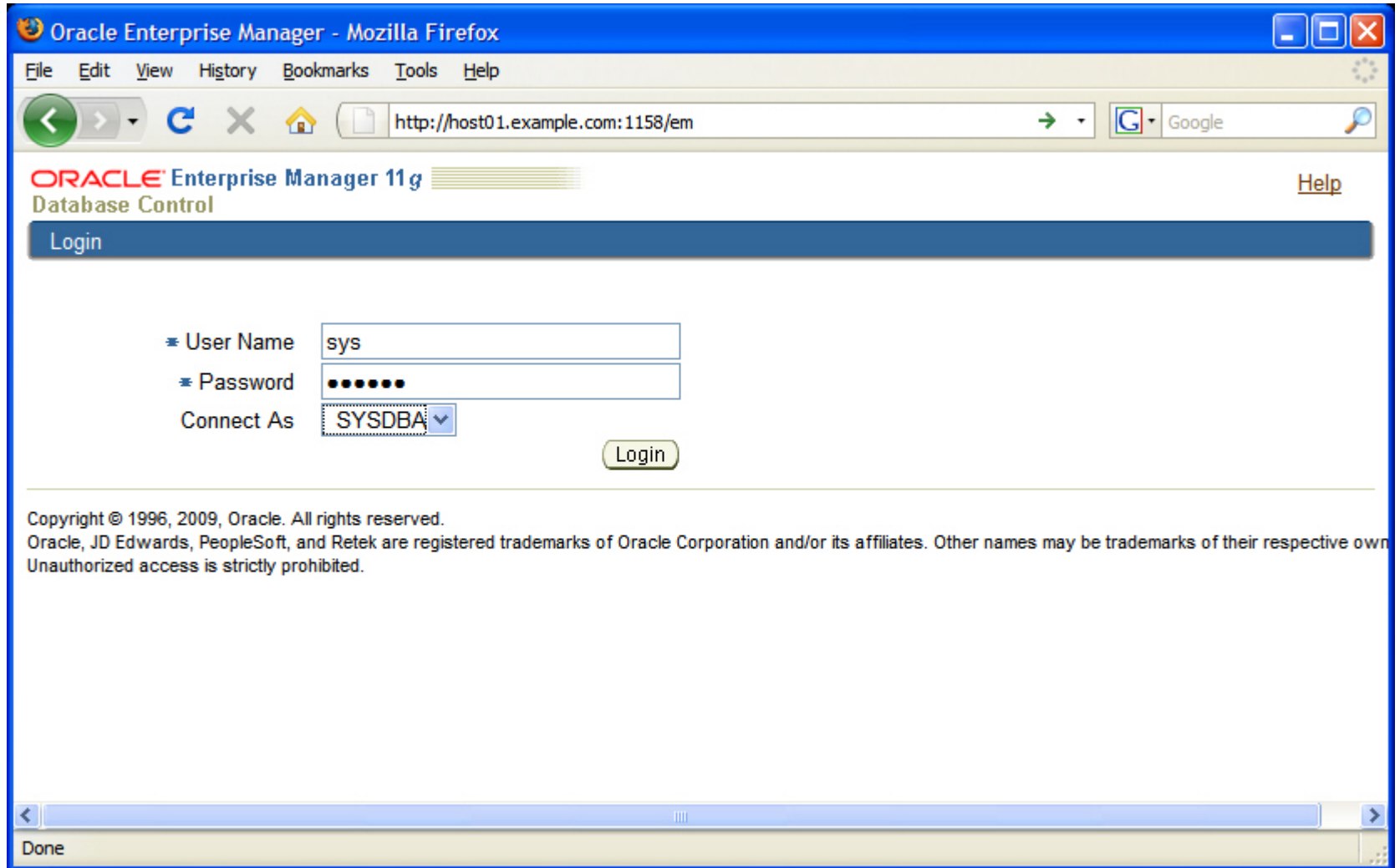


# Starting and Stopping Database Control

```
$ . oraenv
ORACLE_SID = [orcl] ? orcl
The Oracle base for ORACLE_HOME=/u01/app/oracle/product/11.2.0/db_home1
is /u01/app/oracle
$ emctl start dbconsole
Oracle Enterprise Manager 11g Database Control Release 11.2.0.1.0
Copyright (c) 1996, 2009 Oracle Corporation. All rights reserved.
http://host01.example.com:1158/em/console/aboutApplication
Starting Oracle Enterprise Manager 11g Database Control .....started.
-----
Logs are generated in directory
/u01/app/oracle/product/11.2.0/db_home1/host01.example.com_orcl/sysman/
log
```

```
$ emctl stop dbconsole
Oracle Enterprise Manager 11g Database Control Release 11.2.0.1.0
Copyright (c) 1996, 2009 Oracle Corporation. All rights reserved.
https://host01.example.com:1158/em/console/aboutApplication
Stopping Oracle Enterprise Manager 11g Database Control ...
... Stopped.
```

# Oracle Enterprise Manager



# Database Home Page

**ORACLE** Enterprise Manager 11g Database Control

Setup Preferences Help Logout

Database

Logged in As SYS

Database Instance: orcl.example.com

Home Performance Availability Server Schema Data Movement Software and Support

Page Refreshed Jun 18, 2009 11:46:00 PM GMT+07:00 Refresh View Data Automatically (60 sec)

### General

Shutdown Black Out

Status Up

Up Since Jun 18, 2009 5:31:03 AM GMT+07:00

Instance Name orcl

Version 11.2.0.1.0

Host edrsr25p1.us.oracle.com

Listener LISTENER\_edrsr25p1.us.orac...

ASM +ASM\_edrsr25p1.us.oracle.com

[View All Properties](#)

### Host CPU

Load 0.31 Paging 0.05

### Active Sessions

Core Count 1

### SQL Response Time

SQL Response Time (%) 102.90

[Edit Reference Collection](#)

### Diagnostic Summary

|                            |                                    |
|----------------------------|------------------------------------|
| ADDM Findings Period       | 2                                  |
| Start Time                 | Jun 18, 2009 10:00:40 PM GMT+07:00 |
| Alert Log Active Incidents | No ORA- errors                     |

### Space Summary

|                                 |       |
|---------------------------------|-------|
| Database Size (GB)              | 1.448 |
| Problem Tablespaces             | 0     |
| Segment Advisor Recommendations | 0     |
| Policy Violations               | 0     |
| Dump Area Used (%)              | 85    |

### High Availability

|                                |                 |
|--------------------------------|-----------------|
| Console Oracle Restart         | Details Enabled |
| Instance Recovery Time (sec)   | 14              |
| Last Backup                    | n/a             |
| Usable Flash Recovery Area (%) | 95.96           |
| Flashback Database Logging     | Disabled        |

# Other Oracle Tools

Components  
> **SQL\*Plus**  
Init Params  
DB Startup  
DB Shutdown  
Alert Log  
Perf Views

- SQL\*Plus provides an additional interface to your database so that you can:
  - Perform database management operations
  - Execute SQL commands to query, insert, update, and delete data in your database
- SQL Developer:
  - Is a graphical user interface for accessing your instance of Oracle Database
  - Supports development in both SQL and PL/SQL
  - Is available in the default installation of Oracle Database

# Using SQL\*Plus

SQL\*Plus is:

- A command-line tool
- Used interactively or in batch mode

```
$ sqlplus hr
```

```
SQL*Plus: Release 11.2.0.1.0 - Production on Thu Jun 18 05:04:49 2009  
Copyright (c) 1982, 2009, Oracle. All rights reserved.  
Enter Password: *****
```

```
Connected to:
```

```
Oracle Database 11g Enterprise Edition Release 11.2.0.1.0 - Production  
With the Partitioning, Automatic Storage Management, OLAP, Data Mining  
and Real Application Testing options
```

```
SQL> select last_name from employees;
```

```
LAST_NAME
```

```
-----
```

```
Abel
```

```
Ande
```

```
...
```



# Calling SQL\*Plus from a Shell Script

```
$ ./batch_sqlplus.sh
```

```
SQL*Plus: Release 11.2.0.1.0 - Production on Thu Jun 18 05:10:19 2009  
Copyright (c) 1982, 2009, Oracle. All rights reserved.
```

```
Connected to:
```

```
Oracle Database 11g Enterprise Edition Release 11.2.0.1.0 - Production  
With the Partitioning, Automatic Storage Management, OLAP, Data Mining  
and Real Application Testing options
```

```
SQL>
```

```
  COUNT(*)
```

```
-----
```

```
      107
```

```
SQL>
```

```
107 rows updated.
```

```
SQL>
```

```
Commit complete.
```

```
SQL> Disconnected from Oracle Database 11g Enterprise Edition Release  
11.2.0.1.0 - Production
```

```
With the Partitioning, Automatic Storage Management, OLAP, Data Mining  
and Real Application Testing options
```

```
$
```

```
# Name of this file: batch_sqlplus.sh  
# Count employees and give raise.  
sqlplus hr/hr <<EOF  
select count(*) from employees;  
update employees set salary = salary*1.10;  
commit;  
quit  
EOF
```

Output

# Calling a SQL Script from SQL\*Plus

script.sql

```
select * from departments where location_id = 1400;  
quit
```

↓  
Output

```
$ sqlplus hr/hr @script.sql
```

```
SQL*Plus: Release 11.2.0.1.0 - Production on Thu Jun 18 05:13:42 2009  
Copyright (c) 1982, 2009, Oracle. All rights reserved.
```

```
Connected to:
```

```
Oracle Database 11g Enterprise Edition Release 11.2.0.1.0 - Production  
With the Partitioning, Automatic Storage Management, OLAP, Data Mining  
and Real Application Testing options
```

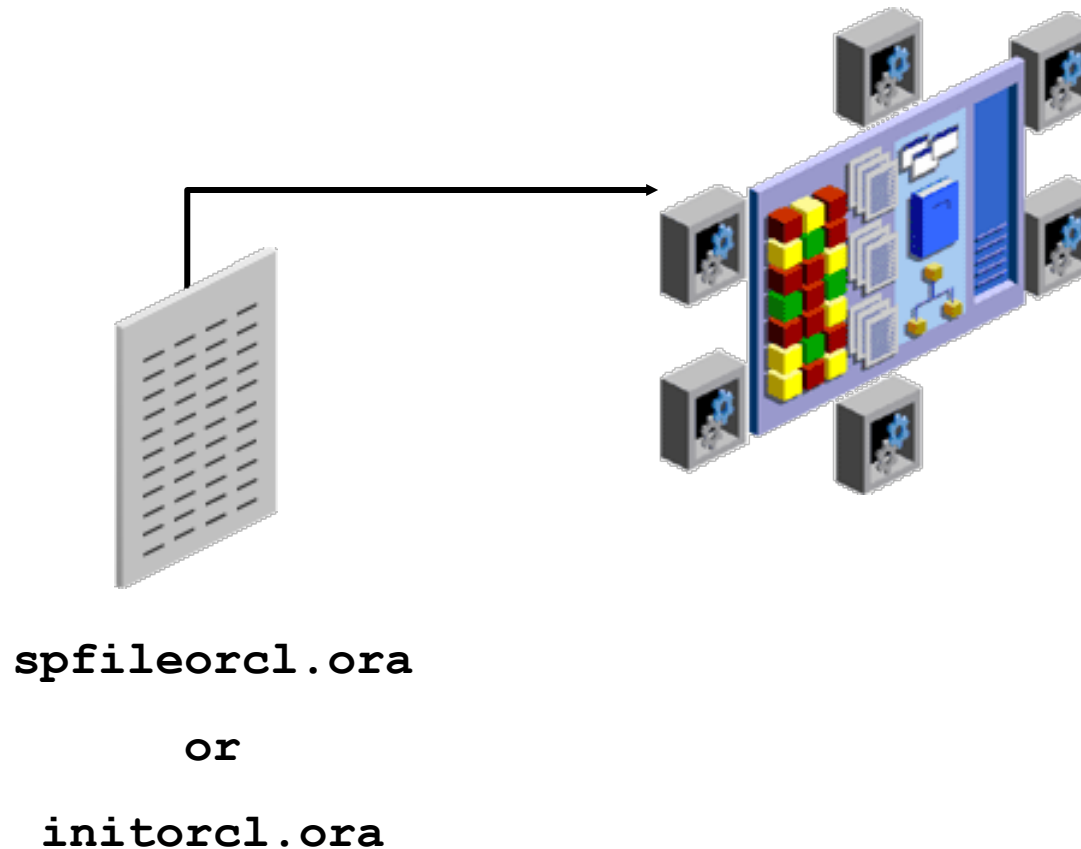
| DEPARTMENT_ID | DEPARTMENT_NAME | MANAGER_ID | LOCATION_ID |
|---------------|-----------------|------------|-------------|
| 60            | IT              | 103        | 1400        |

```
Disconnected from Oracle Database 11g Enterprise Edition Release  
11.2.0.1.0 - Production
```

```
With the Partitioning, Automatic Storage Management, OLAP, Data Mining  
and Real Application Testing options
```

# Initialization Parameter Files

Components  
SQL\*Plus  
> **Init Params**  
DB Startup  
DB Shutdown  
Alert Log  
Perf Views



# Simplified Initialization Parameters

Basic



**CONTROL\_FILES**  
**DB\_BLOCK\_SIZE**  
**PROCESSES**  
**UNDO\_TABLESPACE**  
...

Advanced

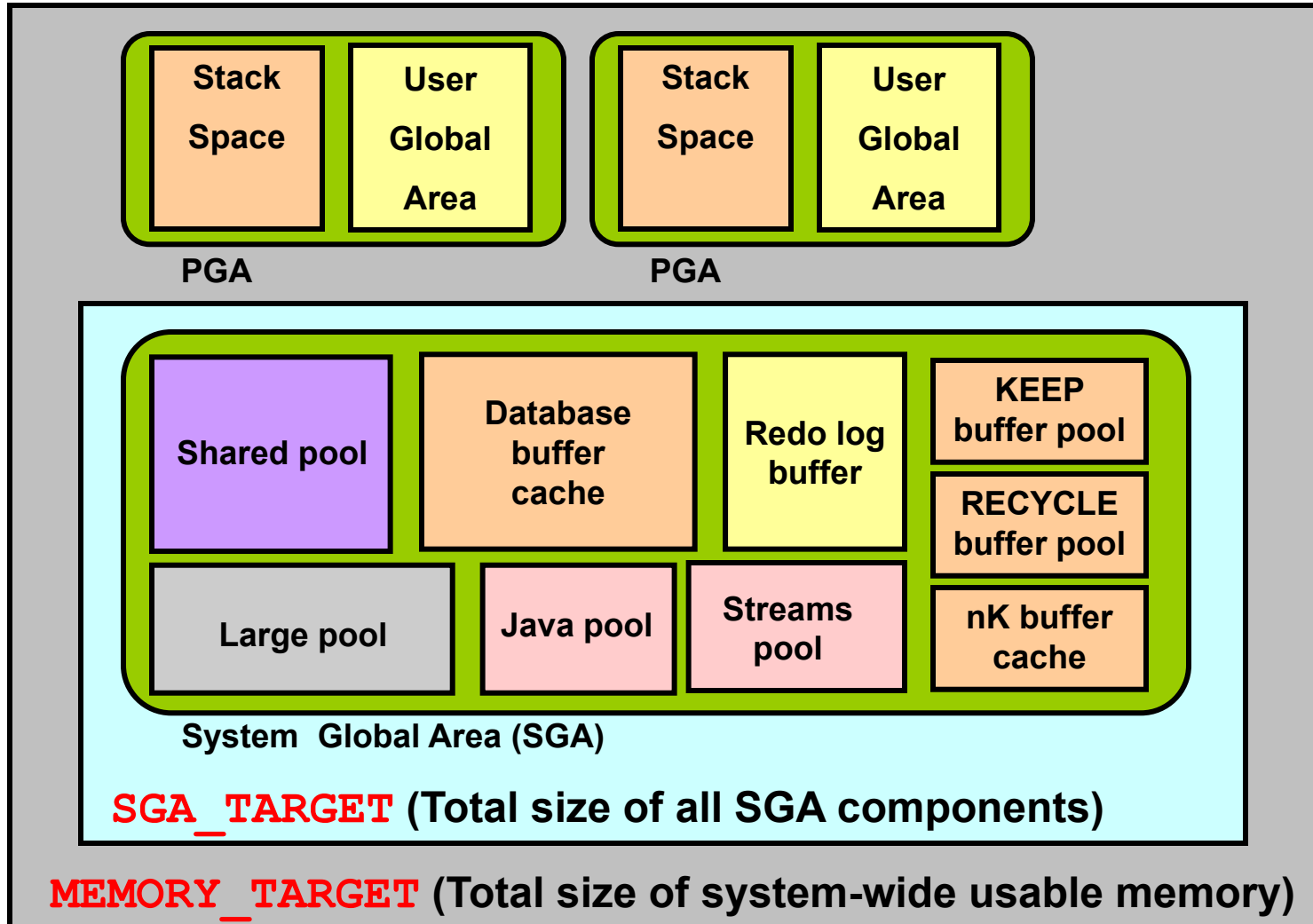


**DB\_CACHE\_SIZE**  
**DB\_FILE\_MULTIBLOCK**  
**\_READ\_COUNT**  
**SHARED\_POOL\_SIZE**  
...

# Initialization Parameters: Examples

| Parameter     | Specifies                                                           |
|---------------|---------------------------------------------------------------------|
| CONTROL_FILES | One or more control file names                                      |
| DB_FILES      | Maximum number of database files                                    |
| PROCESSES     | Maximum number of OS user processes that can simultaneously connect |
| DB_BLOCK_SIZE | Standard database block size used by all tablespaces                |
| DB_CACHE_SIZE | Size of the standard block buffer cache                             |

# Initialization Parameters: Examples



# Initialization Parameters: Examples

| Parameter            | Specifies                                              |
|----------------------|--------------------------------------------------------|
| PGA_AGGREGATE_TARGET | Amount of PGA memory allocated to all server processes |
| SHARED_POOL_SIZE     | Size of shared pool (in bytes)                         |
| UNDO_MANAGEMENT      | Undo space management mode to be used                  |

# Using SQL\*Plus to View Parameters

```
SQL> SELECT name , value FROM V$PARAMETER;
```

| NAME                | VALUE |
|---------------------|-------|
| lock_name_space     | 2     |
| processes           | 150   |
| sessions            | 247   |
| timed_statistics    | TRUE  |
| timed_os_statistics | 0     |
| ...                 |       |

```
SQL>SHOW PARAMETER SHARED_POOL_SIZE
```

| NAME             | TYPE        | VALUE |
|------------------|-------------|-------|
| shared_pool_size | big integer | 0     |

```
SQL> show parameter para
```

| NAME                            | TYPE    | VALUE |
|---------------------------------|---------|-------|
| fast_start_parallel_rollback    | string  | LOW   |
| parallel_adaptive_multi_user    | boolean | TRUE  |
| parallel_automatic_tuning       | boolean | FALSE |
| parallel_execution_message_size | integer | 16384 |
| parallel_instance_group         | string  |       |
| ...                             |         |       |



# Changing Initialization Parameter Values

- Static parameters:
  - Can be changed only in the parameter file
  - Require restarting the instance before taking effect
  - Account for about 110 parameters
- Dynamic parameters:
  - Can be changed while database is online
  - Can be altered at:
    - Session level
    - System level
  - Are valid for duration of session or based on `SCOPE` setting
  - Are changed by using `ALTER SESSION` and `ALTER SYSTEM` commands
  - Account for about 234 parameters

# Changing Parameter Values: Examples

```
SQL> ALTER SESSION  
      SET NLS_DATE_FORMAT = 'mon dd yyyy';
```

Session altered.

```
SQL> SELECT SYSDATE FROM dual;
```

```
SYSDATE  
-----  
jun 18 2009
```

```
SQL> ALTER SYSTEM SET  
SEC_MAX_FAILED_LOGIN_ATTEMPTS=2 COMMENT='Reduce  
from 10 for tighter security.' SCOPE=SPFILE;
```

System altered.

# Quiz

Enterprise Manager Database Control can be used to manage many databases concurrently.

1. True
2. False

# Quiz

The majority of the database parameters are dynamic and can be changed without having to shut down the database instance.

1. True
2. False

# Database Startup and Shutdown: Credentials

Components  
SQL\*Plus  
Init Params  
> **DB Startup**  
DB Shutdown  
Alert Log  
Perf Views

ORACLE® Enterprise Manager 11g Database Control

Setup Preferences Help Logout

Database

Database Instance: orcl.example.com > Logged in As SYS

Cancel OK

### Startup/Shutdown: Specify Host and Target Database Credentials

Specify the following credentials in order to change the status of the database.

#### Host Credentials

Specify the OS user name and password to login to target database machine.

\* Username

\* Password

#### Database Credentials

Specify the credentials for the target database.

To use OS authentication, leave the user name and password fields blank.

\* Username

\* Password

Database

\* Connect As

☐ Save as Preferred Credential

Note that you need to login to the database as SYSDBA or SYSOPER in order to change the status of the database.

Cancel OK

General

Shutdown

or 1

Database Instance

Startup

# Starting Up an Oracle Database Instance

**Database Instance: orcl.example.com**  
Enterprise Manager is not able to connect to the database instance. The state of the components are listed below.  
Page Jun 19, 2009  
Refreshed 1:39:14 AM GMT+07:00  
Refresh

**Database Instance** (1)  
Status Down  
Host edrsr25p1.us.oracle.com  
Port 1521  
SID orcl  
Oracle Home /u01/app/oracle/product/11.2.0/db\_home1  
Details There has been a user-initiated shutdown.

**Host Credentials**  
Specify the OS user name and password  
Username oracle  
Password .....

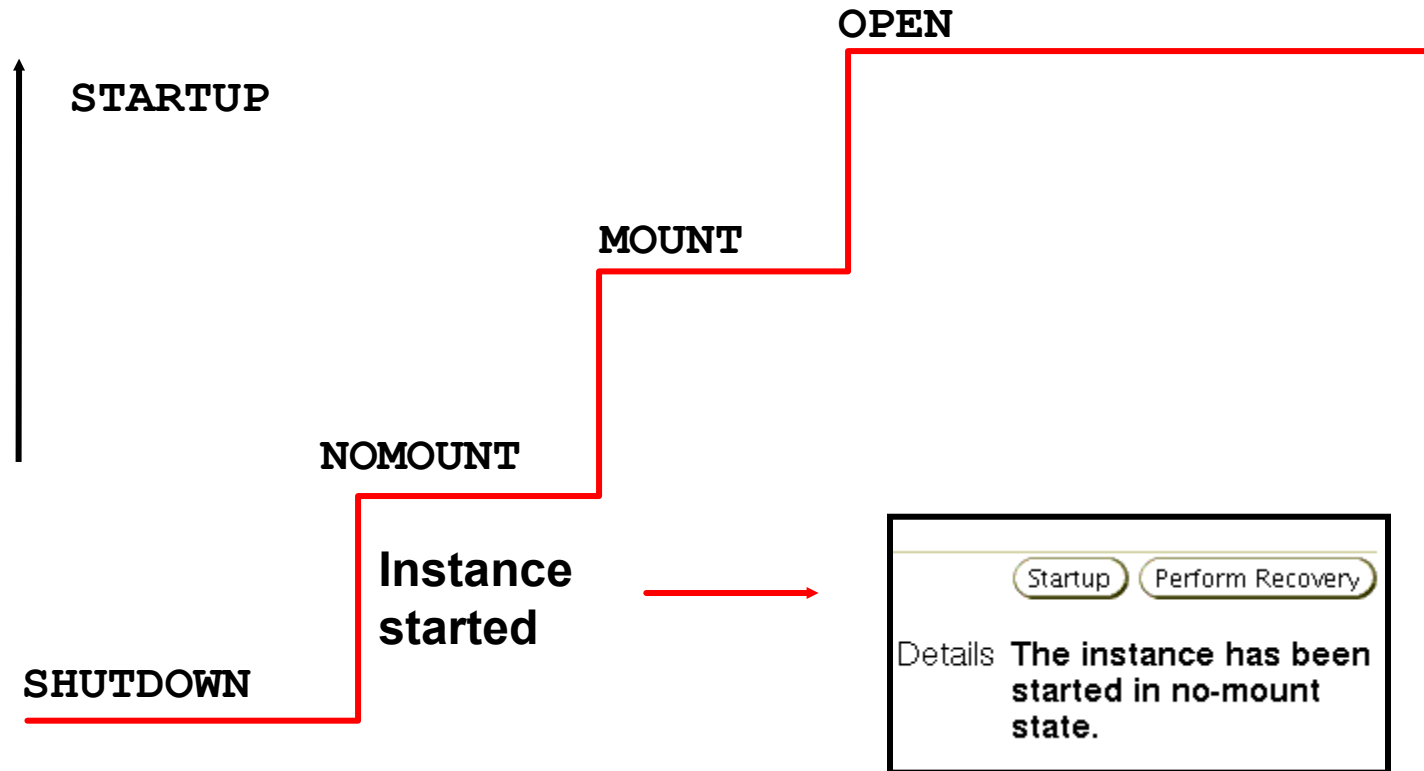
**Database Credentials** (2)  
Specify the credentials for the target

**Select Startup Type** (3)  
This database is registered with Oracle Restart. This enables you to use srvctl utility that comes with Oracle Restart to start this database. Using srvctl will attempt to start the database resource and all other resources on which this database depends (eg: listener, ASM instance etc). Alternatively you may attempt to start the database alone using sqlplus utility. Choose the way in which you want to start the database.  
Select Startup Type  
☒ Start database along with dependent resources  
☐ Start database only  
Cancel OK

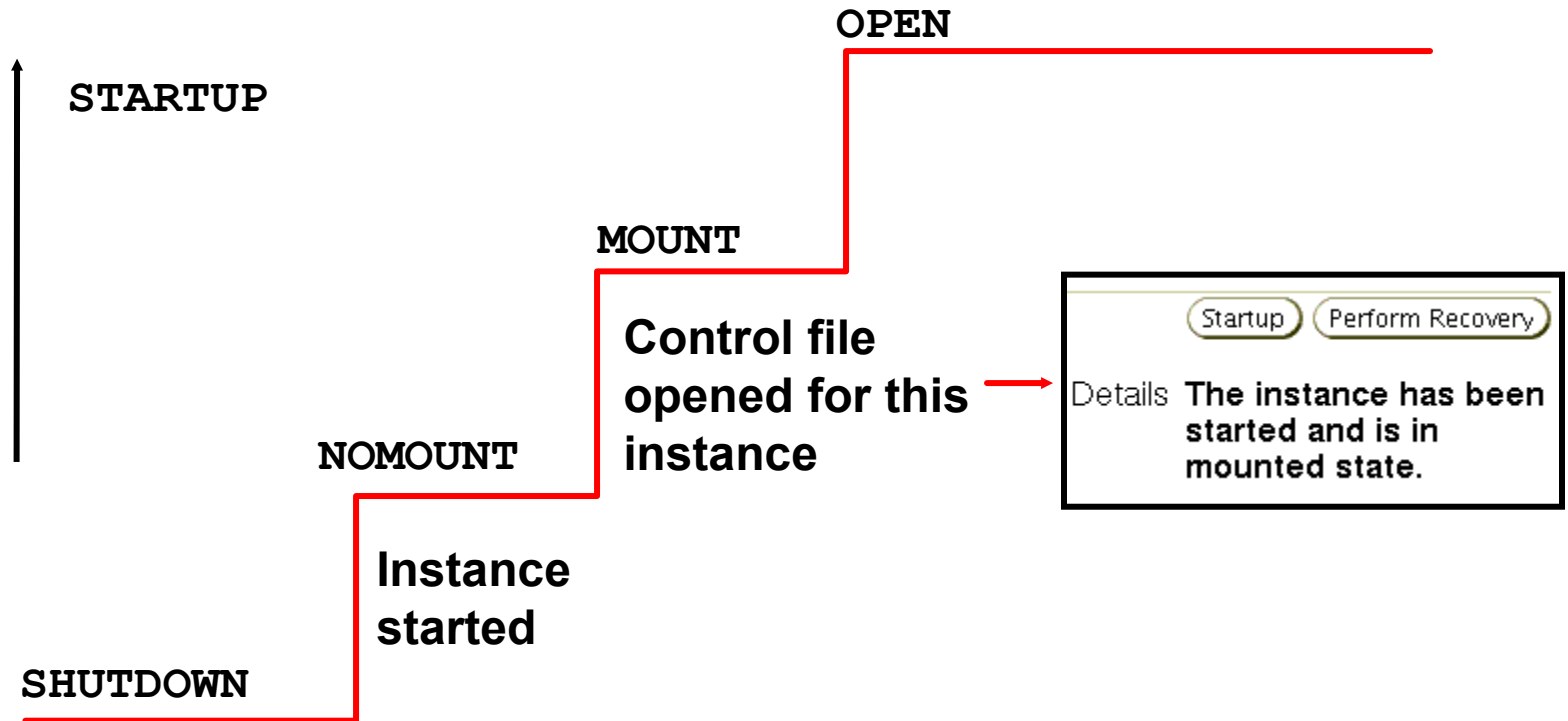
**Startup: Advanced Options** (5)  
**Startup mode**  
☐ Start the database  
☐ Mount the database  
☒ Open the database  
**Other Startup Options**  
☐ Restrict access to database  
☐ Force the instance(s) to start

**Startup/Shutdown: Confirmation** (4)  
Instances orcl  
Operation startup instance(s) in open mode  
Are you sure you want to perform this operation?  
Advanced Options No Yes

# Starting Up an Oracle Database Instance: NOMOUNT

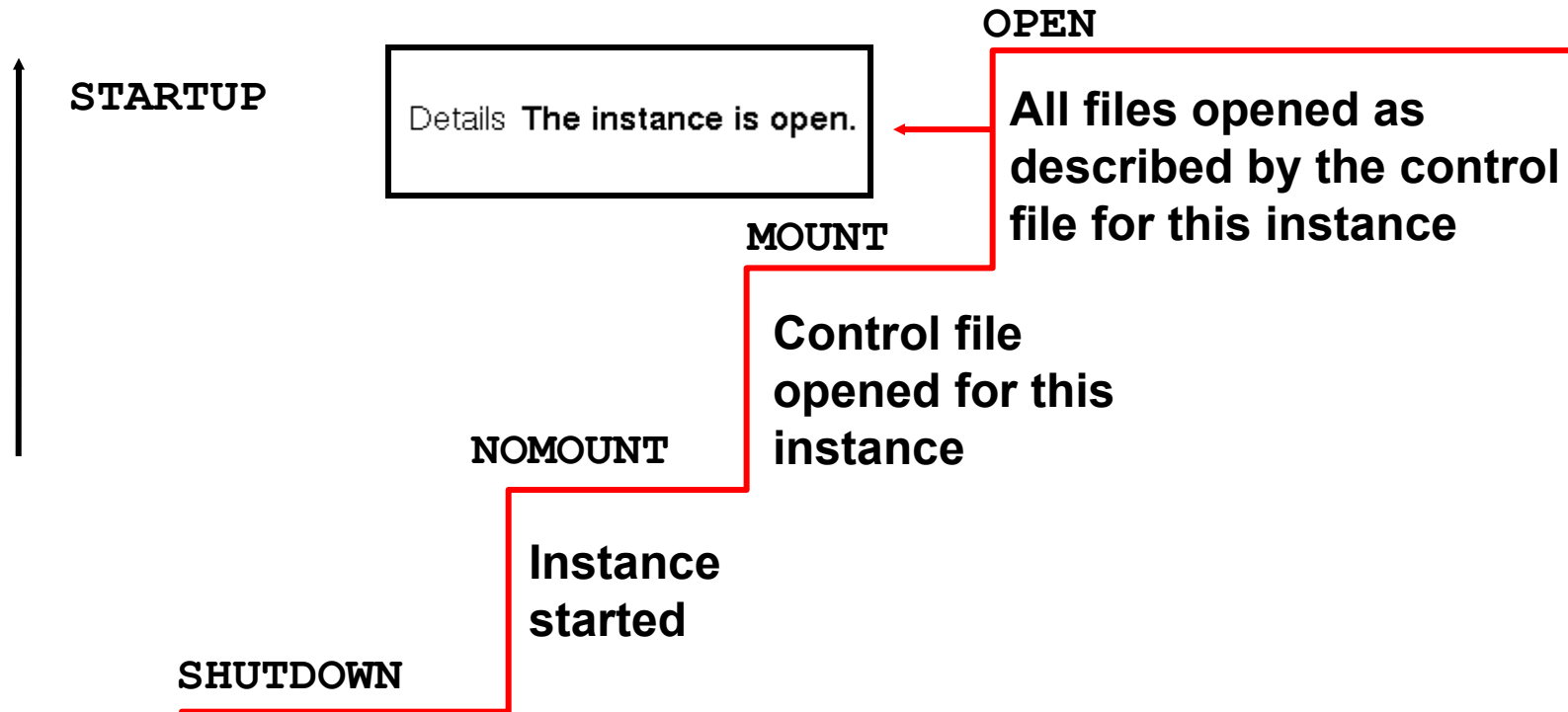


# Starting Up an Oracle Database Instance: MOUNT





# Starting Up an Oracle Database Instance: OPEN



# Startup Options: Examples

- Using the `sqlplus` utility:

```
SQL> startup
```

1

```
SQL> startup nomount
```

2

```
SQL> alter database mount;
```

3

```
SQL> alter database open;
```

4

- Using the `srvctl` utility with Oracle Restart

```
$ srvctl start database -d orcl -o mount
```

# Shutting Down an Oracle Database Instance

**General** 1

↑

Status **Up**

Up Since **Jun 19, 2009 2:02:00 AM GMT+07:00**

Instance Name **orcl**

Version **11.2.0.1.0**

Host **edrsr25p1.us.oracle.com**

Listener **LISTENER\_edrsr25p1.us.orac...**

ASM **+ASM\_edrsr25p1.us.oracle.com**

[View All Properties](#)

**Shutdown** **Black Out**

**Startup/Shutdown: Specify Host and Target Database Credentials**

Specify the following credentials in order to change the status of the database.

**Host Credentials**

Specify the OS user name and password to login to target database machine.

\* Username **oracle**

\* Password **.....**

**Database Credentials**

Specify the credentials for the target database.  
To use OS authentication, leave the user name and password fields blank.

\* Username **sys**

\* Password **.....**

Database **orcl.example.com**

\* Connect As **SYSDBA** ▼ 2

**Startup/Shutdown: Advanced Shutdown Option**

Specify the shutdown mode

☐ Normal [Browse Sessions](#)

☒ **Wait for all currently connected users to disconnect from the database**

☐ Transactional

Disconnect all connected users after transactions have completed

☒ **Immediate**

Rollback active transactions and disconnect all connected users

☐ Abort

Instantaneous shutdown by aborting the database instance 4

**Startup/Shutdown: Confirmation** 3

Current Status **open**

Operation **shutdown immediate**

Are you sure you want to perform this operation?

**Show SQL** **Advanced Options** **No** **Yes**

**Cancel** **OK**

# Shutdown Modes

| Shutdown Modes                       | A  | I   | T   | N   |
|--------------------------------------|----|-----|-----|-----|
| Allows new connections               | No | No  | No  | No  |
| Waits until current sessions end     | No | No  | No  | Yes |
| Waits until current transactions end | No | No  | Yes | Yes |
| Forces a checkpoint and closes files | No | Yes | Yes | Yes |

## Shutdown modes:

- A = ABORT
- I = IMMEDIATE
- T = TRANSACTIONAL
- N = NORMAL

# Shutdown Options

On the way down:

- Uncommitted changes rolled back, for IMMEDIATE
- Database buffer cache written to data files
- Resources released

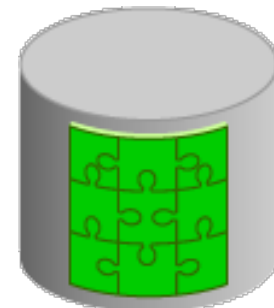
During:

SHUTDOWN  
NORMAL  
or  
SHUTDOWN  
TRANSACTIONAL  
or  
SHUTDOWN  
IMMEDIATE

On the way up:

- No instance recovery

**Consistent database**

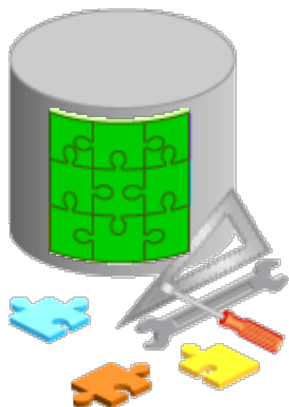


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# Shutdown Options

On the way down:

- Modified buffers not written to data files
- Uncommitted changes not rolled back



**During:**

SHUTDOWN ABORT  
or  
Instance failure  
or  
STARTUP FORCE

On the way up:

- Online redo log files used to reapply changes
- Undo segments used to roll back uncommitted changes
- Resources released

**Inconsistent database**

# Shutdown Options: Examples

- Using SQL\*Plus:

```
SQL> shutdown
```

1

```
SQL> shutdown transactional
```

2

```
SQL> shutdown immediate
```

3

```
SQL> shutdown abort
```

4

- Using the SRVCTL utility with Oracle Restart

```
$ srvctl stop database -d orcl -o abort
```

# Viewing the Alert Log

Database Home page > Related Links region > Alert Log Content

Components  
SQL\*Plus  
Init Params  
DB Startup  
DB Shutdown  
> **Alert Log**  
Perf Views

| View Entries <input type="button" value="Last 50"/> <input type="button" value="Go"/> <input type="button" value="Search"/> |              |       |             |               |                                 |                                                                                           |
|-----------------------------------------------------------------------------------------------------------------------------|--------------|-------|-------------|---------------|---------------------------------|-------------------------------------------------------------------------------------------|
| Previous 1-25 of 50 Next 25                                                                                                 |              |       |             |               |                                 |                                                                                           |
| Timestamp                                                                                                                   | Type         | Level | Incident ID | Group         | Message ID                      | Message Text                                                                              |
| Jun 19, 2009 10:00:16 PM GMT+07:00                                                                                          | NOTIFICATION | 16    |             | sqltune       | kesaiTuneSqlDrv:5067:3456118459 | End automatic SQL Tuning Advisor run for special tuning task "SYS_AUTO_SQL_TUNING_TASK"   |
| Jun 19, 2009 10:00:03 PM GMT+07:00                                                                                          | NOTIFICATION | 16    |             | sqltune       | kesaiTuneSqlDrv:4555:2579917519 | Begin automatic SQL Tuning Advisor run for special tuning task "SYS_AUTO_SQL_TUNING_TASK" |
| Jun 19, 2009 10:00:00 PM GMT+07:00                                                                                          | NOTIFICATION | 16    |             | process start | ksbrdp:3833:3697353022          | VKRM started with pid=24, OS id=7929                                                      |
| Jun 19, 2009 10:00:00 PM GMT+07:00                                                                                          | NOTIFICATION | 16    |             | process start | ksbs1p_real:2253:2371767696     | Starting background process VKRM                                                          |
| Jun 19, 2009 2:07:22 AM GMT+07:00                                                                                           | NOTIFICATION | 16    |             | process start | ksbrdp:3833:3697353022          | SMCO started with pid=23, OS id=30582                                                     |
| Jun 19, 2009 2:07:22 AM GMT+07:00                                                                                           | NOTIFICATION | 16    |             | process start | ksbs1p_real:2253:2371767696     | Starting background process SMCO                                                          |
| Jun 19, 2009 2:02:26 AM GMT+07:00                                                                                           | NOTIFICATION | 16    |             | process start | ksbrdp:3833:3697353022          | CJQ0 started with pid=33, OS id=29846                                                     |

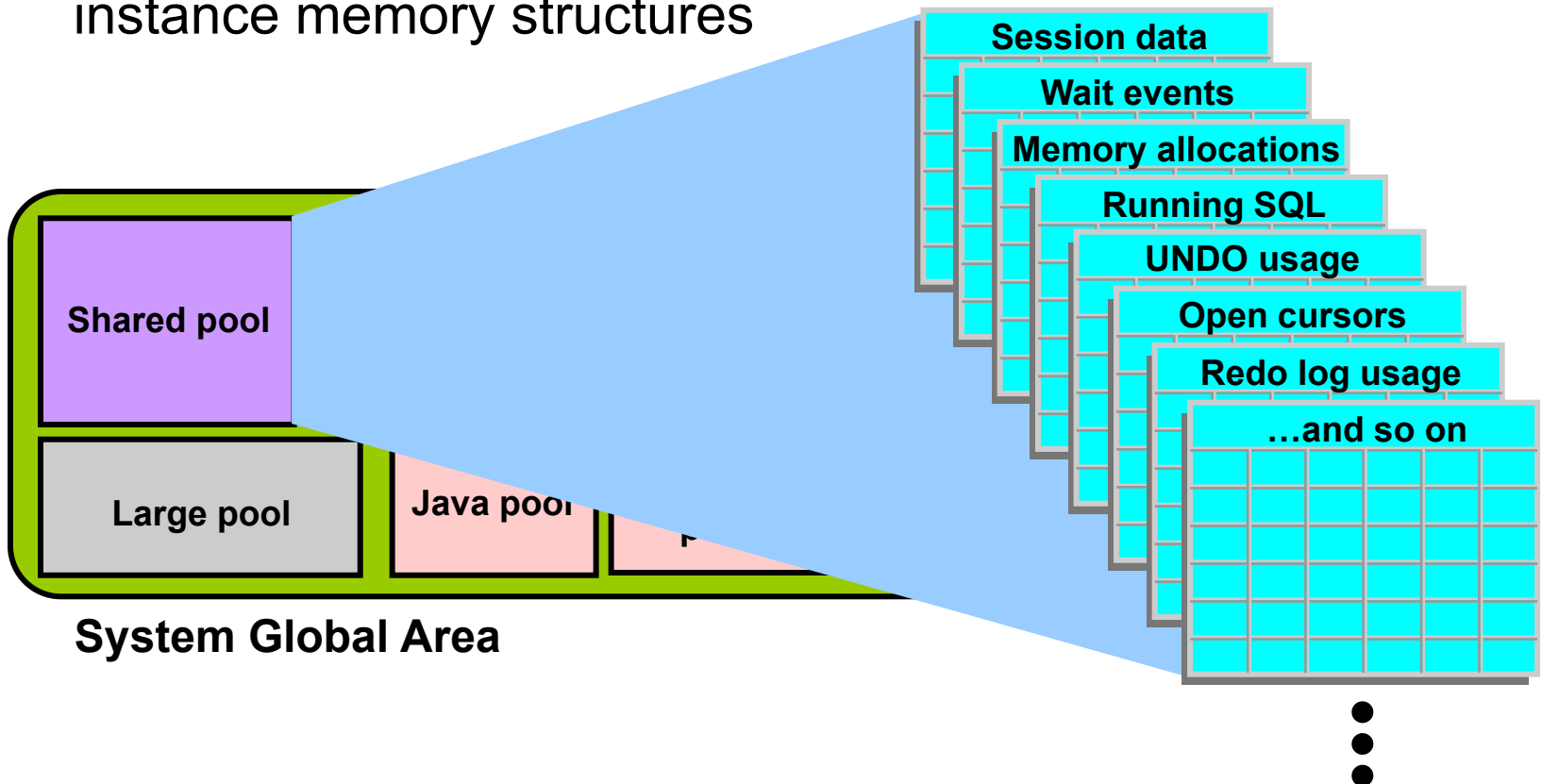


# Using Trace Files

- Each server and background process can write to an associated trace file.
- Error information is written to the corresponding trace file.
- Automatic diagnostic repository (ADR)
  - Is a systemwide central tracing and logging repository
  - Stores database diagnostic data such as:
    - Traces
    - Alert log
    - Health monitor reports

# Dynamic Performance Views

Provide access to information about changing states of the instance memory structures



# Dynamic Performance Views: Usage Examples

1

```
SQL> SELECT sql_text, executions FROM v$sql  
WHERE cpu_time > 200000;
```

2

```
SQL> SELECT * FROM v$session WHERE machine =  
'EDRSR9P1' and logon_time > SYSDATE - 1;
```

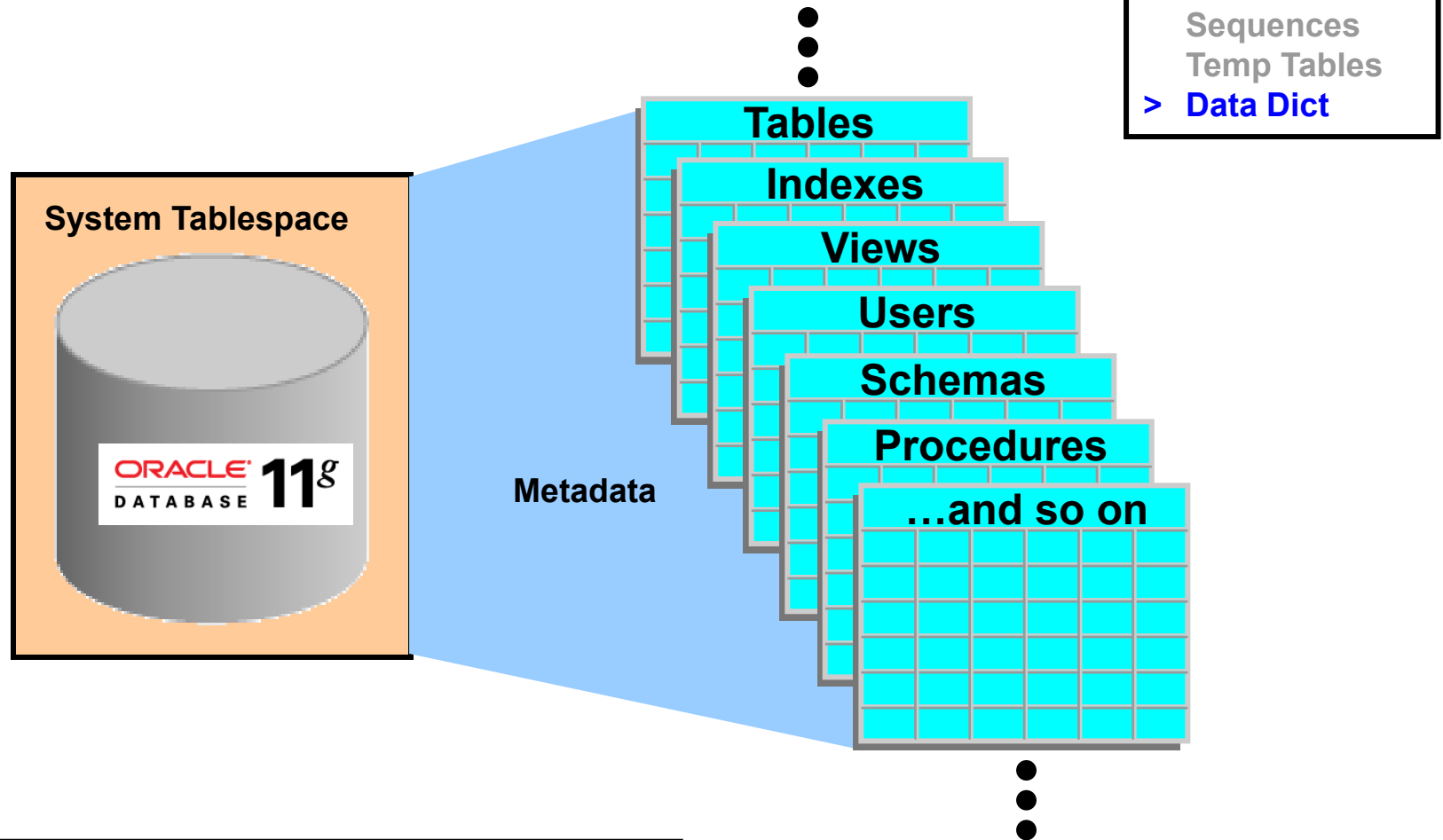
3

```
SQL> SELECT sid, ctime FROM v$lock  
WHERE block > 0;
```

# Dynamic Performance Views: Considerations

- These views are owned by the `SYS` user.
- Different views are available at different times:
  - The instance has been started.
  - The database is mounted.
  - The database is open.
- You can query `V$FIXED_TABLE` to see all the view names.
- These views are often referred to as “v-dollar views.”
- Read consistency is not guaranteed on these views because the data is dynamic.

# Data Dictionary: Overview



```
SELECT * FROM dictionary;
```

# Data Dictionary Views

|       | Who Can Query | Contents                                       | Subset of  | Notes                                                                                                                   |
|-------|---------------|------------------------------------------------|------------|-------------------------------------------------------------------------------------------------------------------------|
| DBA_  | DBA           | Everything                                     | N/A        | May have additional columns meant for DBA use only                                                                      |
| ALL_  | Everyone      | Everything that the user has privileges to see | DBA_ views | Includes user's own objects and other objects the user has been granted privileges to see                               |
| USER_ | Everyone      | Everything that the user owns                  | ALL_ views | Is usually the same as ALL_ except for the missing OWNER column (Some views have abbreviated names as PUBLIC synonyms.) |

# Data Dictionary: Usage Examples

1

```
SELECT table_name, tablespace_name  
FROM user_tables;
```

2

```
SELECT sequence_name, min_value, max_value,  
increment_by  
FROM all_sequences  
WHERE sequence_owner IN ('MDSYS', 'XDB');
```

3

```
SELECT USERNAME, ACCOUNT_STATUS  
FROM dba_users  
WHERE ACCOUNT_STATUS = 'OPEN';
```

4

```
DESCRIBE dba_indexes
```

# Quiz

When using Oracle Restart, the server control utility (`srvctl`) must be used instead of SQL\*Plus to start and stop a database instance.

1. True
2. False



# Quiz

Which data dictionary view can be used to find the names of all tables in the database?

1. USER\_TABLES
2. ALL\_TABLES
3. DBA\_TABLES
4. ANY\_TABLES

# Summary

In this lesson, you should have learned how to:

- Start and stop the Oracle database and components
- Use Oracle Enterprise Manager
- Access a database with SQL\*Plus
- Modify database initialization parameters
- Describe the stages of database startup
- Describe database shutdown options
- View the alert log
- Access dynamic performance views

# **Practice 4 Overview: Managing the Oracle Instance**

This practice covers the following topics:

- Navigating in Enterprise Manager
- Viewing and modifying initialization parameters
- Stopping and starting the database instance
- Viewing the alert log
- Connecting to the database by using SQL\*Plus